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STUDIES ON A CHILD POPULATION

II. RETESTS ON THE ADVANCED OTIS AND STANFORD-BINET SCALES, WITH NOTES ON THE USE OF A SHORTENED BINET SCALE

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INTRODUCTION

THE first paper of this series dealt with the definition and ascertainment of a population of school children. In addition the results were given of an application of the Advanced Otis scale. The group was composed of all children whose homes were within the boundaries of the City of Bath on 27 July 1934 and whose dates of birth fell between 1 September 1921 and 31 August 1925 inclusive. The completeness of the ascertainment was assessed in the first paper; since then some improvement has been effected: it is intended to give particulars of the state of the ascertainment in the next paper.

Observations are being made upon the children composing the group and upon several special subsamples selected from it. Most of this work involves the measurement of the association of other factors with intelligence. Before proceeding to describe the results, it is necessary to examine for our own group certain features of the functioning of the intelligence tests that we are using. The present paper therefore deals with this examination.

In the first place details are given of a retest of a sample of the children on the Advanced Otis scale. It is desirable that the test-retest association should be estimated for the

group that is being studied, for it does not follow that retests carried out by others on groups differently ascertained would yield applicable results. The Index of Brightness deduced from the application of the Otis scale will be used for a variety of purposes and a knowledge of the test-retest association for the group will enable appropriate adjustments to be made.

In much of the work on the group the Stanford-Binet scale is being used, so it was felt to be necessary to retest a subsample of the children using this scale. In this way the relation between Otis and Binet scales could be defined for our own material. In addition, as the Binet scale is universally accepted as the standard mental test, it is desirable that its relation to others used in studies such as this should be investigated. It was also desirable to specify the distribution of our population in terms of the Binet scale.

Ultimately quite a large proportion of the children comprising the group (together with the brothers and sisters of many of them) will be individually tested. When so many individual tests have to be given any device that can shorten the time taken is of great service. Accordingly details are given of the error introduced by the use of a shortened Binet scale that we are using for some of the children who are being individually tested.

A RETEST ON THE ADVANCED OTIS SCALE

(1) *General note*

Much attention has been devoted by educational psychologists to the "validity" and "reliability" of mental tests. When use is made of the results of such tests these considerations are of considerable practical importance, for the magnitude and accuracy of many estimates may be appreciably altered by using appropriate adjustments.

An ordinary mental test is not a single measurement. It is a battery or scale of tests, the result being an average or total value. The first question therefore is the extent to which the various items are a good sample of the indefinitely large population of tests that might be applied to measure the subject's response within a given field of mental activity. Estimates of variability in this sense may be obtained by examining the covariance of two halves of the scale, or by examining that of odd and even items, or of the results of applying two separate scales designed to test the same abilities. Such investigations belong to the field of educational psychology and are of course of importance in the designing of batteries of tests.

If, however, the results of mental testing are being used for such a purpose, for example, as the estimation of the sib-sib correlation of intelligence, it is another aspect of the variability of results that is of immediate importance. The measurement is that of the subject's response to given stimuli and will vary from day to day for a variety of obvious reasons. Ideally we should like to know the sampling error to be attached to the result obtained on a single occasion. Clearly an estimate may be obtained from the association of test and retest. Anastasi (1934) draws the distinction between the two aspects of

"reliability" as follows "...individual's responses to the objectively same stimuli on different occasions; in the latter with the individual's response to *different stimuli* selected to sample the same general field of activity".

The contrast may be illustrated by a simple analogy. Results are available for the number of living full sibs of a group of children all aged 10 years. The observer desires to measure the association between fertility and family income. He may, however, consider that fertility should be measured by the total number of pregnancies of a woman during the reproductive period. His data are thus faulty, for he has no record of stillbirths, early deaths, children of previous marriages, etc. Further, in some cases the reproductive period of the mother is not over. Similarly his data for family income might refer to a given week, whereas what he would really like to know is the average family income over a period of years. If, however, the results of other detailed investigations were available, it might be possible to estimate "fertility" and "income", as he would like to define them, from the data he has and also to determine the sampling error of his estimates. This corresponds to the first consideration discussed above in relation to mental testing.

On the other hand, the observer might be content to consider simply the data he has, without reference to his ideal criteria. He will then only have to take into account the accuracy of the information supplied to him. Clearly, errors in determining and recording the number of sibs, and the family income for a given week, could be estimated very simply by repeating the observations. This corresponds to the estimation of test-retest association on an intelligence scale.

In the case of investigations such as ours, it is difficult to avoid the conclusion that it is best to take into account simply the test-retest variability. This can be readily determined for the group that is being studied, an important point in view of the great differences in methods of ascertainment in different studies. We have chosen a particular test, the Advanced Otis, that has been widely applied, and is generally accepted as a good group measure. For our purposes we may accept it as the fixed series of objective stimuli to which the child makes his measured response.

The measurement of the test-retest association, however, presents certain difficulties in the case of a mental test. If the interval is short there may be a practice effect. If the interval is long the practice effect is reduced but an appreciable amount of growth has taken place. Actually, however, the question of growth does not seem to raise serious difficulty owing to the peculiar relation of original score to increment described in this paper. This is at least true as regards our own material. The interval between test and retest, one year, should reduce the practice effect to fairly small limits, so that it seems reasonable to accept the test-retest data in estimating variability of response, though of course, unlike retests of physical measurements, it cannot supply a criterion that is at all perfect.

The long interval seems to be a positive advantage in another way, for, ideally, we should like to know the error of estimate of a response drawn from a population of responses

given by an individual exhibiting various emotional and other differences over a long period of time.

(2) *Selection of the subsamples*

It was decided to test the subsample after the lapse of a year, plus or minus 7 days; in this way the calculations relating to age at test were greatly simplified. The ideal plan would have been to have retested a random sample, but for obvious reasons the practical difficulties involved were insuperable; not only would every school have had to be revisited in order to test a portion of every class, but all the children at, for example, a senior school, who had moved up from various junior schools, would have been tested originally on a variety of dates.

As the specifications of the original complete group were known, it was possible to select representative samples as regards the original test results. The original complete group comprised four age groups and of these the oldest three were tested in 1934 (apart from some private schools) as was described in the first paper of this series. The oldest age group was originally tested at senior, secondary schools, etc., so that the great majority were still at the same schools in the summer of 1935; similarly the great majority of the third age group were still at junior schools: the second age group was not suitable for sampling as the children had moved up from the junior schools at the end of the summer term of 1934.

Two representative samples were therefore chosen, one from the third age group (averaging about 10 years 3 months at original test) and one from the oldest age group (averaging about 12 years 5 months).

In the case of the younger sample a proportion of the junior schools was chosen, so that the children within the selected age group gave a distribution corresponding approximately to the distribution of scores for the whole age group. Lists were prepared showing the children of this age group originally tested at each of the schools, which were then visited within 7 days of the dates on which the tests were carried out in 1934. The numbers were sufficient to provide a group of at least 250, after allowing for the omission of absentees. It was found, however, that owing to the tendency for some of the cleverest children to go on to secondary schools before attaining the usual age, there was a deficiency of very clever children. This was allowed for by retesting a few very clever children only at one or two other junior schools, and also by including a small group tested at nearly the same date in 1934, who were at the girls' secondary school in 1935.

The selection of a subsample of similar size from the oldest age group was simpler. By choosing one secondary school, the central school and a proportion of the senior schools, a sample was readily selected that corresponded approximately in original scores to the distribution of the entire age group.

Table I gives a comparison between the mean score and the variance of the two subsamples and the expected figures based on the whole group. The latter values are taken from Tables VIII and IX of the first paper.

Table I

Subsample	No.	Mean age yrs. mos.	Mean score	Norm at that age	Variance	Expected variance
Younger	258	10 3	61.6	66	1066	1076
Older	274	12 5	103.1	102	1335	1368

The children of the younger subsample fell rather below the expected figure in mean score. This, however, was true of most of the group from which they were drawn, as is shown in Table VI of the previous paper.

A more detailed comparison may be made. The actual distribution of scores in broad groups is given below in Table II. This is compared with the distribution of scores of children of the whole group of ages 9 years 10 months to 10 years 9 months, and 11 years 11 months to 12 years 10 months, respectively. These figures are taken from the Appendix table in the first paper.

Table II. *Younger subsample*

Percentage frequencies: scores

	No.	0-19	20-39	40-59	60-79	80-99	100-119	120-139	140-159	160-179
Whole group	741	9.9	15.5	18.8	19.6	20.5	10.1	4.6	0.8	0.3
Subsample	258	9.3	19.0	23.6	17.8	15.5	11.2	2.7	0.8	0.0

Older subsample

Percentage frequencies: scores

	No.	0-19	20-39	40-59	60-79	80-99	100-119	120-139	140-159	160-179	180-199
Whole group	914	3.0	3.2	5.9	12.8	20.0	23.4	16.5	10.7	3.7	0.8
Subsample	274	1.5	5.1	6.9	10.2	19.0	21.9	18.6	13.1	3.6	0.0

It will be seen that the two subsamples are fairly closely representative in scores at first test of the group from which they were drawn. The subsamples were, of course, not necessarily representative in any other way, for example, no private school children were included.

(3) *The relation between first and second scores*

Appendix tables I and II show the first and second scores of the children composing the younger and older groups respectively. The scores were grouped with a class interval of 5 units, i.e. approximately one-seventh of the average standard deviation. The loss of information due to grouping should thus be very small.

Tables III and IV show the calculations necessary for the investigation of the relationship.

The correlations, 0.9345 and 0.9574 respectively, indicate a satisfactorily high test-retest association. It will also be noted that the residual variance between arrays after eliminating linear regression is not significant in either case. The relationship between first and second

Table III. *Younger group: first and second scores* x = first score; y = second score

x	No.	$\bar{y}_p$	Y_p	$n_p (\bar{y}_p - Y_p)^2$
0-4	6	15.3333	27.7681	928
5-9	6	34.5000	32.9052	15
10-14	9	32.0000	38.0424	329
15-19	3	47.0000	43.1795	44
20-24	9	52.5556	48.3167	162
25-29	13	48.1538	53.4538	365
30-34	12	63.2500	58.5909	261
35-39	15	62.0000	63.7281	45
40-44	11	71.0909	68.8652	54
45-49	16	74.1875	74.0023	1
50-54	12	78.2500	79.1395	9
55-59	22	86.3182	84.2766	92
60-64	12	92.0000	89.4137	80
65-69	10	98.5000	94.5509	156
70-74	8	108.8750	99.6880	675
75-79	16	105.1250	104.8251	1
80-84	12	108.2500	109.9623	35
85-89	16	112.6250	115.0994	98
90-94	8	124.5000	120.2365	145
95-99	4	114.5000	125.3737	473
100-104	9	138.6667	130.5108	599
105-109	7	138.4286	135.6479	54
110-114	3	138.6667	140.7851	13
115-119	10	140.5000	145.9222	294
120-124	1	142.0000	151.0594	82
125-129	3	152.0000	156.1965	53
135-139	3	165.3333	166.4708	4
145-149	2	157.0000	176.7450	780
	258	—	—	5847

$\bar{x} = 61.61240$, $S(x - \bar{x})^2 = 274011$, $\bar{y} = 89.01550$, $S(y - \bar{y})^2 = 335652$,
 $Sx(y - \bar{y}) = 281527$, $S\{n_p(\bar{y}_p - \bar{y})^2\} = 295095$, $b\{Sx(y - \bar{y})\} = 289248$.

Regression of second score on first score = 1.027427.

Regression of first score on second score = 0.838745.

Correlation coefficient = 0.934481.

Analysis of variance

Variation of second score	Degrees of freedom	Sum of squares	Mean square	$\frac{1}{2} \log_e$
Linear regression	1	289248	289248	—
Remainder between arrays	26	5847	224.9	2.7079
Within arrays	230	40557	176.3	2.5861
Total	257	335652	—	—

 $z = 0.1218$

5% point approximately 0.216.

Table IV. *Older group: first and second scores* x = first score; y = second score

x	No.	$\bar{y}_p$	Y_p	$n_p (\bar{y}_p - Y_p)^2$
0-4	1	22.0000	23.0488	1
10-14	1	32.0000	32.9648	.1
15-19	2	27.0000	37.9229	239
20-24	2	37.0000	42.8809	69
25-29	5	45.0000	47.8390	40
30-34	6	57.8333	52.7970	152
35-39	1	42.0000	57.7551	248
40-44	6	66.1667	62.7131	72
45-49	4	63.2500	67.6711	78
50-54	3	63.6667	72.6292	241
55-59	6	78.6667	77.5872	7
60-64	9	92.5556	82.5453	902
65-69	6	93.6667	87.5033	228
70-74	4	87.0000	92.4614	119
75-79	9	96.4444	97.4194	9
80-84	7	94.8571	102.3774	396
85-89	12	100.7500	107.3355	520
90-94	15	116.6667	112.2935	287
95-99	18	114.7778	117.2516	110
100-104	16	125.1250	122.2096	136
105-109	14	128.0714	127.1677	11
110-114	17	131.1177	132.1257	17
115-119	13	141.6154	137.0837	267
120-124	13	138.9231	142.0418	126
125-129	17	150.5294	146.9998	212
130-134	12	149.0833	151.9579	99
135-139	9	162.5556	156.9159	286
140-144	4	157.0000	161.8740	95
145-149	16	166.0625	166.8320	9
150-154	10	170.5000	171.7900	17
155-159	6	177.8333	176.7481	7
160-164	6	178.6667	181.7061	55
165-169	2	187.0000	186.6642	0
175-179	2	189.5000	196.5803	100
	274	—	—	5159

 $\bar{x} = 103.05839$, $S(x - \bar{x})^2 = 364443$, $\bar{y} = 123.25912$, $S(y - \bar{y})^2 = 390991$, $Sx(y - \bar{y}) = 361385$, $S\{n_p(\bar{y}_p - \bar{y})^2\} = 363511$, $b\{Sx(y - \bar{y})\} = 358352$.

Regression of second score on first score = 0.991609.

Regression of first score on second score = 0.924280.

Correlation coefficient = 0.957353.

Analysis of variance

Variation of second score	Degrees of freedom	Sum of squares	Mean square	$\frac{1}{2} \log_e$
Linear regression	1	358352	358352	—
Remainder between arrays	32	5159	161.2	2.5413
Within arrays	240	27480	114.5	2.3703
Total	273	390991	—	—

 $z = 0.1710$

5% point approximately 0.200.

scores is thus of the simplest description and it will be sufficient in making adjustments to rely entirely on the linear regression coefficients or upon the correlation coefficient.

The transformation of r to z permits a test to be made of the significance of the difference between the correlations in the two subsamples. Table VB in Fisher's *Statistical Methods for Research Workers* gives corresponding values of r and z . The difference is 2.5 times its standard error and must be considered significant; the test-retest association of the Advanced Otis scale is therefore higher in the case of the older children.

	r	z	$n'-3$	$\frac{1}{n'-3}$	$(n'-3) z^*$
Younger group	0.9345	1.6931	255	0.003922	431.28
Older group	0.95735	1.9131	271	0.003690	517.96
Difference	—	0.2200 $\pm$	0.0872	0.007612	949.24

* Allowing for the small positive bias in the value of z (Fisher, 5th ed. p. 192).

As the two subsamples represent the extreme distribution of ages at test of the complete group, a single figure based on the combination of the two values may be considered a reasonable estimate to use in making adjustments. Using the figure given above the weighted average value of z is 1.8046 ± 0.0436 , and the corresponding value of r is 0.9473. The chance is 5 per cent that the value of r lies outside 0.9554 and 0.9378; the chance is 1 per cent that the value of r lies outside 0.9577 and 0.9344. If the allowance is made in the form of an error variance the figures necessary for the calculation are given in Tables III and IV.

(4) *The relation between first and second Indexes of Brightness*

In the first paper of the series a description was given of an equalization for age by the transformation of scores into I.B.'s. It is to be expected that the association between first and second I.B.'s will be somewhat less close than between first and second scores, for extremely large numbers would be required to establish an almost perfect age correction.

Table V. *Younger group: first and second I.B.'s*

$$\bar{x} = 95.81783, \quad S(x - \bar{x})^2 = 305614, \quad \bar{y} = 104.82946, \quad S(y - \bar{y})^2 = 336635, \quad Sx(y - \bar{y}) = 297763.$$

$$\begin{aligned} \text{Regression of second I.B. on first I.B.} &= 0.974309. \\ \text{Regression of first I.B. on second I.B.} &= 0.884529. \\ \text{Correlation coefficient} &= 0.928334. \end{aligned}$$

The comparison of the first and second I.B.'s can only be made in the case of the younger group, because the children comprising the older group exceeded in age at second test the ages covered by our table of norms. A summary of the result for the younger group is

given in Table V. The correlation coefficient of 0.9345 for first and second scores has fallen to 0.9283 if first and second I.B.'s are compared.

	r	z	$n'-3$	$\frac{1}{n'-3}$
Scores	0.9345	1.6931	255	0.003922
I.B.'s	0.9283	1.6457	255	0.003922
Difference	—	0.0474 $\pm$	0.0886	0.007844

The difference is not significant, and its small size may be taken as one indication that the method of age correction described in the first paper is working reasonably well.

It will be noted that there is an apparent increase in average I.B. of about 9 units. This is probably to be attributed to a practice effect and is discussed further in a later section.

(5) *A further analysis of the increment of score during a year*

The results of the retesting permit some further examination to be made of the increment of score during a year.

Probably the most important question that can be answered is whether there is a relationship between the amount of the score at first test and the amount of the subsequent increment. It would not be unreasonable to expect that the increment would, on the average, be greater in the case of children securing high scores originally, as compared with children securing low scores.

Tables VI and VII present the results for the two groups. As in these tables, and in the remaining tables in this section, no regression is significant, nor any remainder between arrays, it is not worth while setting out the fitting in detail; accordingly summaries only are given. It is perhaps remarkable that in both groups the increment was apparently

Table VI. *Younger group*

$$\begin{aligned} x &= \text{first score; } y = \text{increment; } n = 258 \\ \bar{x} &= 61.61240, \quad S(x - \bar{x})^2 = 274011, \quad \bar{y} = 27.29070, \quad S(y - \bar{y})^2 = 44553, \\ Sx(y - \bar{y}) &= 7729, \quad S\{n_s(\bar{y}_s - \bar{y})^2\} = 5394, \quad b\{Sx(y - \bar{y})\} = 218. \end{aligned}$$

Regression of increment on first score = 0.028207.

Regression of first score on increment = 0.173480.

Correlation coefficient = 0.069953.

Analysis of variance

Variation of increment	Degrees of freedom	Sum of squares	Mean square	$\frac{1}{2} \log_e$
Linear regression	1	218	218.0	2.6923
Remainder between arrays	26	5176	199.1	2.6469
Within arrays	230	39159	170.3	2.5688
Total	257	44553	—	—

Neither difference significant.

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Table VII. *Older group*

x = first score; y = increment; $n = 274$
 $\bar{x} = 103.05839$, $S(x - \bar{x})^2 = 364443$, $\bar{y} = 20.24818$, $S(y - \bar{y})^2 = 32709$,
 $Sx(y - \bar{y}) = -967$, $S\{n_p(\bar{y}_p - \bar{y})^2\} = 5174$, $b\{Sx(y - \bar{y})\} = 2.6$.
 Regression of increment on first score = -0.002653 .
 Regression of first score on increment = -0.029563 .
 Correlation coefficient = -0.008857 .

Analysis of variance

Variation of increment	Degrees of freedom	Sum of squares	Mean square	$\frac{1}{2} \log_e$
Linear regression	1	2.6	2.6	—
Remainder between arrays	32	5171	161.6	2.5425
Within arrays	240	27535	114.7	2.3712
Total	273	32709	—	—

Neither difference significant.

quite independent of the original score, but there seems to be no room for doubt as the numbers are relatively large. A comparison with Table IX in our first paper shows that the variance of second scores was approximately that to be expected at the ages then attained. It is clear that if there is no association between original score and increment, the variance of second score will be equal to the variance of first score plus the variance of the increment. This is illustrated by our figures, if allowance is made for the small non-significant product term.

An examination of the correlation table, in the case of the younger subsample especially, suggested that although mean increment was constant in relation to original score, it was possible that the increment became less variable as original score increased. To test this point a further regression was examined, that of the square of the increment on original score. A summary is presented in Table VIII. It will be seen that there is no significant

Table VIII. *Younger group*

x = first score; y = square of increment; $n = 258$
 $\bar{x} = 61.61240$, $S(x - \bar{x})^2 = 274011$, $\bar{y} = 917.46899$, $S(y - \bar{y})^2 = 146661270$,
 $Sx(y - \bar{y}) = 35397$, $S\{n_p(\bar{y}_p - \bar{y})^2\} = 15727063$, $b\{Sx(y - \bar{y})\} = 4573$.
 Regression of square of increment on first score = 0.129180 .
 Regression of first score on square of increment = 0.000241 .
 Correlation coefficient = 0.005584 .

Analysis of variance

Variation of square of increment	Degrees of freedom	Sum of squares	Mean square	$\frac{1}{2} \log_e$
Linear regression	1	4573	4573	—
Remainder between arrays	26	15722490	604711	6.6563
Within arrays	230	130934207	569279	6.6261
Total	257	146661270	—	—

Neither difference significant.

association, so that the variability of the increment also must be considered independent of original score, as far as our numbers are adequate for testing this point.

Age at test is another factor that might be expected to influence the amount of the increment. A comparison of the younger and older samples shows that the increment is appreciably less in the older group.

	Mean age at original test yrs. mos.	Increment
Younger group	10 3	27.2907 ± 0.8197
Older group	12 5	20.2482 ± 0.6613
Difference	2 2	$7.0425 \pm 1.0532 (6.7X)$

The difference is highly significant, so that with an age difference of this size there is no doubt that the older children add on fewer marks during the course of a year.

Analysis of the relation within the groups between age at original test and increment is much less satisfactory. The range in age in each case is only just over a year. Further, the samples were not representative as regards age at test; as was previously explained, it was necessary, in the case of the younger group, to test some very clever children at extra schools. These children tended to be young. The regressions of original score on age, given in Tables XI and XII, show that these are much too small, being non-significant on our numbers in the younger group, and negligible in the older group.

Tables IX and X give summaries of the relationship between age at test and increment for the two groups. Bearing in mind the limitations, mentioned above, it is unlikely that there can be any close association, and investigations with numbers such as ours, and confined to a narrow age range, are unlikely to disclose any relationship.

Table IX. *Younger group*

$$x = \text{age at first test (above 9 years); } y = \text{increment; } n = 258$$

$$\bar{x} = 15.015504, \quad S(x - \bar{x})^2 = 3094, \quad \bar{y} = 27.290698, \quad S(y - \bar{y})^2 = 44553,$$

$$Sx(y - \bar{y}) = -886, \quad S\{n_p(\bar{y}_p - \bar{y})^2\} = 3597, \quad b\{Sx(y - \bar{y})\} = 254.$$

Regression of increment on age = -0.286419 .

Regression of age on increment = -0.019890 .

Correlation coefficient = -0.075478 .

Analysis of variance

Variation of increment	Degrees of freedom	Sum of squares	Mean square	$\frac{1}{2} \log_e$
Linear regression	1	254	254	2.7687
Remainder between arrays	13	3343	257.2	2.7750
Within arrays	243	40956	168.5	2.5635
Total	251	44553	—	—

Neither difference significant.

Table X. *Older group*

x = age at first test (above 11 years); y = increment; $n = 274$
 $\bar{x} = 16.613139$, $S(x - \bar{x})^2 = 3369$, $\bar{y} = 20.248175$, $S(y - \bar{y})^2 = 32709$,
 $Sx(y - \bar{y}) = 724$, $S\{n_p(\bar{y}_p - \bar{y})^2\} = 1643$, $b\{Sx(y - \bar{y})\} = 156$.

Regression of increment on age = 0.214992.

Regression of age on increment = 0.022144.

Correlation coefficient = 0.068998.

Analysis of variance

Variation of increment	Degrees of freedom	Sum of squares	Mean square	$\frac{1}{2} \log_e$
Linear regression	1	156	156	2.5250
Remainder between arrays	12	1487	123.9	2.4097
Within arrays	261	31066	119.0	2.3896
Total	274	32709	—	—

Neither difference significant.

This analysis may be completed by giving a summary of the linear relationships between the variables, original score, increment and age at original test, for both subsamples, and in addition giving the important partial regressions. This is done in Tables XI and XII. No simple regression is significant, not even those of first score on age; the partial regressions of increment on age and first score, at fixed first score and at fixed age respectively, are not significant either.

Table XI. *Younger group*

x = age at first test (above 9 years); y = first score; z = increment
 $\bar{x} = 15.015504$, $\bar{y} = 61.612403$, $\bar{z} = 27.290698$, $n = 258$.

Sums of squares and products of deviations:

	x	y	z
x	3093.94	2731.55	-886.163
y	—	274011	7729.07
z	—	—	44553.2

Regressions

y on $x = 0.882872 \pm 0.585583$, x on $y = 0.009969 \pm 0.006612$.

z on $x = -0.286419 \pm 0.236496$, x on $z = -0.019890 \pm 0.016423$.

z on $y = 0.028207 \pm 0.025140$, y on $z = 0.173480 \pm 0.154618$.

Sums of squares and products of deviations at fixed age:

	y	z
y	271600	8511.44
z	—	44299.4

Regression of increment on first score at fixed age = 0.031338 ± 0.025215 .

Table XI (*continued*)

Sums of squares and products of deviations at fixed first score:

	<i>x</i>	<i>z</i>
<i>x</i>	3066.71	-963.212
<i>z</i>	—	44335.2

Regression of increment on age at fixed first score = -0.314087 ± 0.237291 .Table XII. *Older group*

x = age at first test (above 11 years); *y* = first score; *z* = increment
 $\bar{x} = 16.613139$, $\bar{y} = 103.058394$, $\bar{z} = 20.248175$, $n = 274$.

Sums of squares and products of deviations:

	<i>x</i>	<i>y</i>	<i>z</i>
<i>x</i>	3368.99	227.191	724.307
<i>y</i>	—	364443	-966.970
<i>z</i>	—	—	32709.2

Regressions

y on *x* = 0.067436 ± 0.630625 , *x* on *y* = 0.000623 ± 0.005830 .
z on *x* = 0.214992 ± 0.188479 , *x* on *z* = 0.022144 ± 0.019413 .
z on *y* = -0.002653 ± 0.018164 , *y* on *z* = -0.029563 ± 0.202385 .

Sums of squares and products of deviations at fixed age:

	<i>y</i>	<i>z</i>
<i>y</i>	364428	-1015.81
<i>z</i>	—	32553.4

Regression of increment on first score at fixed age = -0.002787 ± 0.018155 .

Sums of squares and products of deviations at fixed first score:

	<i>x</i>	<i>z</i>
<i>x</i>	3368.85	724.910
<i>z</i>	—	32706.6

Regression of increment on age at fixed first score = 0.215180 ± 0.188823 .(6) *Discussion and summary*

The object in view in administering the retest was to secure figures for the test-retest association shown by the Advanced Otis scale as applied by ourselves to the group of children upon which we are carrying out various investigations. In the case of ordinary

physical measurements, errors of estimation are usually so small that any correction for departures from reliability could have no appreciable effect upon say a sib correlation. This is not the case with mental tests. A correlation coefficient of 0.9 between original test results and the results on repetition would be considered reasonable for many of the tests commonly employed; this, of course, corresponds to an error variance of 10.5 per cent and it is clear that the value of a sib correlation, for example, would be appreciably affected.

The children retested were of two groups, each with an age range of approximately 1 year and with mean ages of 10 years 3 months and 12 years 5 months. They were representative, as regards original score, of those portions of the complete group from which they were drawn. The numbers were 258 and 274 respectively.

The correlations were satisfactorily high, being 0.9345 and 0.9574 respectively. The difference is significant, but as these children represent the extreme range of ages at test of the complete group, and as the difference is not very large, it seems permissible to use an averaged value of

$$0.9473.$$

The 5 per cent points are 0.9554 and 0.9378 and the 1 per cent points 0.9577 and 0.9344, so that the coefficient may be regarded as having been determined with some accuracy. The figures given in the preceding sections of this paper can thus be used for making appropriate adjustments.

Although it was not the purpose of the retesting, it seemed worth while to present the data we have accumulated on the characteristics of the increment of score after the lapse of 1 year. Briefly, it is shown that in both groups the amount of the increment is quite independent of the score at original test, a somewhat surprising result. The variability of the increment too is independent of the size of original score. It is also shown that, within the limits of age within the two groups, and bearing in mind certain features in their selection, it is not possible to demonstrate any relation between the amount of the increment and age at original test. The increment over 1 year does however tend to drop as age at original test is increased, and this can be demonstrated with numbers such as our own, if the difference in age is sufficiently great. The increment in the case of the younger group was 27 marks and in the older group 20 marks, the difference being amply significant.

The amount of the increment in both groups was large, much larger than would have been anticipated on the basis of the regression of score on age given in the first paper of this series for children up to the age of 12. In the younger group the expected mean increment would be about 18 points. It is possible that the regression of score on age for children up to the age of 12 was under-estimated, although, as far as we know, it proved to be higher than that given in other studies. The results of the Stanford-Binet retesting do suggest a slight under-estimation. There is little doubt that there was a practice effect, although it may not have a value as great as 9 points.

The question of norms for the Advanced Otis scale is discussed at the end of the section dealing with the Binet retesting (p. 36).

The increment of 20 points in the case of the older children is also high. It is quite possible that in the first paper we considerably under-estimated the regression of score on age in children over the age of 12. On the other hand, if there is a pronounced practice effect, it is not unreasonable to expect that it would be still more pronounced in older children.

One speculation may be worth recording. It is perhaps conceivable that the practice effect should differ in some systematic way according to the intelligence of the child. For example, the very dull child receiving originally very few marks might fail to profit from the experience of a year before. Similarly the very intelligent child would comprehend the nature of the tasks so thoroughly, that the previous experience could not add to the efficiency with which the second test was completed. On the average it might be expected that the child of more moderate performance would profit most. Given some such systematic relationship, it might be argued that the apparent lack of association between increment and original score was due to a masking of a real association, the practice effect tending to equalize the increments. It is difficult, however, to imagine that the practice effect could so neatly and precisely neutralize a real association; we are of the opinion therefore that this explanation of the lack of association between increment and original score can be dismissed.

For the full investigation of this and other points much more elaborate experiments would clearly have to be carried out. We felt, however, that it was desirable to present such information as we had incidentally obtained.

A RETEST ON THE STANFORD-BINET SCALE

(1) *Selection of the subsamples*

Much of the work that is being carried out on children drawn from the complete group includes individual testing, and in each such case a Stanford-Binet test is being given. It was most desirable therefore that the relationship between the results of this test and of the Advanced Otis should be investigated for our own material.

Unlike the retests on the Otis scale, there was in this instance no objection to retesting a random sample. The question of the time interval between the two tests could not be considered and there was no objection to revisiting every school, as this had to be done at various times for other purposes.

It was decided therefore to select children at random for retesting by choosing all the children from the complete group whose dates of birth fell within certain limits. As we have previously described, the first three age groups were all group tested together. The group testing of the fourth age group was postponed for a year so that the children of this group corresponded in age at test to those of the third age group. A subsample for individual testing was accordingly chosen from each of the first three age groups. The children included were those born in January in 1922, 1923 and 1924. The age group 1921-2 was, however,

so much larger than the others, reflecting as it did the high post-war birth-rate, that it was decided roughly to equalize the three groups by including children born from 13 to 31 January only in 1922.

The numbers of children in the complete group that fell into the three subsamples were as follows:

No.	Tested
71	65
62	60
70	67
203	192

It is disappointing that as many as eleven children escaped individual testing. In most cases this was due to the children having left the city. In a few cases in the oldest age group, children were not available when particular schools were being visited and had attained the leaving age before an opportunity had arisen for giving the test. The Otis I.B.'s of the eleven children were as follows: 46, 86, 98, 99, 118, 123, 125, 137, 139, 139, 147; mean 114.27. The chance that a random sample of eleven should have a mean differing by this amount from the mean of the entire group (as given in the first paper) is more than 1 in 6. There is no indication, therefore, that the loss is not a random one, and it is extremely unlikely that the omission of these children could have any appreciable effect upon the results.

Full Binet tests were given, the testing being continued upwards until a complete year of tests was failed and downwards until a complete year of tests was passed. In no case were less than 3 years of tests given. It would no doubt have been a good plan to have retested the children on the Stanford-Binet scale at a short and fixed time interval after the Otis test. This was quite impracticable. The variation in age at Otis test was, however, small as regards the majority of the children in each group, though the total range was wide, chiefly owing to absentees who were tested at later dates. Similarly, the Binet testing was, for the most part, carried out during a comparatively short period, so that again the majority of ages fell within relatively narrow limits. The interval between Otis and Binet tests was fairly constant for most of the children in each group. A long period, however, elapsed between the tests, about a year and a half in the two younger groups, but less in the oldest. This was because other urgent work had to be completed first, before it was possible to find time for this retesting.

For these reasons it proved impossible to make any adequate examination of Otis scores and Binet mental ages. This investigation is therefore confined to an analysis of Otis I.B.'s and Binet I.Q.'s.

The distribution of chronological ages at both tests is shown in Table XIII.

Table XIII. *Chronological ages at test of three random groups*

Age at Otis test (years and months)	No.	Age at Binet test (years and months)	No.
Children born January 1922			
12.5 -12.6	51	13.0 -13.4	56
12.7 -12.10	8	13.5 -13.8	5
12.11-13.3	6	13.9 -14.2	4
Children born January 1923			
11.0 -11.4	16	12.1 -12.3	7
11.5	33	12.4 -12.9	6
11.6 -11.7	6	12.10-12.11	39
11.8 -12.2	5	13.0 -13.4	8
Children born January 1924			
10.0 -10.4	21	10.4 -10.5	4
10.5	29	10.9 -11.5	3
10.6 -10.8	6	11.10-11.11	47
10.9 -11.2	11	12.0 -12.5	13

(2) *The relation between Otis Index of Brightness and Stanford-Binet Intelligence Quotient*

The full results are shown in Appendix table III. As a first step the results were analysed separately for each of the three groups. It was soon apparent that the differences between the regressions and correlation coefficients were not significant and that the regressions were linear. Accordingly it seems best to present in brief summary form the data for each group separately, to demonstrate the lack of significant difference between the groups, and then to show the fitting for the pooled results.

Table XIV. *Regression of Binet I.Q. on Otis I.B. (ungrouped data)*

Group	No.	Sums of squares and products of deviations		Variation due to regression	Regression coefficient
		(Otis I.B.) ²	(Otis I.B. × Binet I.Q.)		
1	65	83006.2	31441.7	11909.7	0.3787874
2	60	80488.9	33078.6	13594.4	0.4109712
3	67	82320.7	30583.3	11362.1	0.3715142
		245815.8	95103.6	36866.2	—

Regression coefficient based on the three groups taken together = 0.3868897.

	Degrees of freedom	Sum of squares
Variation due to three regressions independently	3	36866.2
Variation due to general regression	1	36794.6
Difference due to heterogeneity of regressions	2	71.6

Mean square due to heterogeneity = 35.8.

This value may be compared with a residual variance within arrays for the total group of 52.2 (see Table XVI). It is thus non-significant.

Table XV. *Regression of Otis I.B. on Binet I.Q. (ungrouped data)*

Group	No.	Sums of squares and products of deviations		Variation due to regression	Regression coefficient
		(Binet I.Q.) ²	(Binet I.Q. × Otis I.B.)		
1	65	15173·8	31441·7	65150·7	2·072111
2	60	15865·0	33078·6	68969·0	2·085005
3	67	14843·1	30583·3	63015·1	2·060443
		45881·9	95103·6	197134·8	—

Regression coefficient based on the three groups taken together = 2·072791.

	Degrees of freedom	Sum of squares
Variation due to three regressions independently	3	197134·8
Variation due to general regression	1	197129·9
Difference due to heterogeneity of regressions	2	4·9

Mean square due to heterogeneity = 2·5.

This value may be compared with a residual variance within arrays for the total group of 271·6 (see Table XVII). It is thus non-significant.

There is thus no reason to suppose that the relation of Otis I.B. and Binet I.Q. differs for the three groups as regards the regressions. As would be anticipated a single correlation coefficient is also adequate. The individual correlation coefficients were: group 1, 0·8859; group 2, 0·9257; group 3, 0·8749; and the adjusted weighted average value was 0·8954.

The mean I.Q.'s at 100 I.B. may also be investigated. They proved to be: group 1, $96·5457 \pm 1·9121$; group 2, $98·5037 \pm 2·1173$; group 3, $101·1461 \pm 1·8332$. The standard errors include the small increments due to the fixing of Otis I.B., as the variance of the mean at fixed I.B. is increased by the square of the difference between observed mean I.B. and 100 I.B., multiplied by the variance of the regression coefficient.

It is possible therefore that as age increases mean I.Q. falls in relation to mean I.B. This may be due to a defect in our norms but it is equally possible that it results from the mechanism of determining the Binet I.Q. at the higher chronological ages.

In view of the great convenience of using a single value in each case throughout the whole group, it seems best to pool the results for the three subsamples. The pooled results (ungrouped figures) are as follows:

No. in group	= 192
Mean I.B.	= 99·88021
Variance	= 1294·567
Mean I.Q.	= 98·70833
Variance	= 241·4328

Regression of I.Q. on I.B.	=	0.383173
Standard error	=	0.014451
Regression of I.B. on I.Q.	=	2.054582
Standard error	=	0.077484
Correlation coefficient	=	0.887277

Tables XVI and XVII show the fitting of the two regression lines. Both regressions were linear.

Table XVI. *Regression of Binet I.Q. on Otis I.B.*

$x = \text{Otis I.B.}; y = \text{Binet I.Q.}$

x	No.	$\bar{y}_p$	Y_p	$n_p (\bar{y}_p - Y_p)^2$
10-19	1	62.0000	66.7153	22
20-29	6	71.0000	70.4864	2
30-39	5	72.8000	74.2574	11
40-49	5	80.6000	78.0285	33
50-59	12	84.7500	81.7995	104
60-69	12	85.2500	85.5706	1
70-79	16	89.1875	89.3416	0
80-89	15	92.0000	93.1126	19
90-99	25	95.2400	96.8837	68
100-109	14	100.1429	100.6547	4
110-119	21	104.2857	104.4258	0
120-129	15	109.6000	108.1968	30
130-139	20	113.3000	111.9678	35
140-149	10	112.7000	115.7389	92
150-159	6	122.5000	119.5099	54
160-169	5	121.4000	123.2810	18
170-179	3	132.0000	127.0520	73
180-189	1	125.0000	130.8231	34
	192	—	—	600

$$\bar{x} = 99.65625, \quad S(x - \bar{x})^2 = 252795, \quad \bar{y} = 98.82813, \quad S(y - \bar{y})^2 = 45633, \\ Sx(y - \bar{y}) = 95330, \quad S\{n_p(\bar{y}_p - \bar{y})^2\} = 36549, \quad b\{Sx(y - \bar{y})\} = 35949, \quad b = 0.377104.$$

Analysis of variance

Variation of Binet I.Q.	Degrees of freedom	Sum of squares	Mean square
Linear regression	1	35949	—
Remainder between arrays	16	600	37.50
Within arrays	174	9084	52.21
Total	191	45633	—

Mean square of remainder between arrays less than mean square within arrays.

Table XVII. *Regression of Otis I.B. on Binet I.Q.* $x = \text{Binet I.Q.}; y = \text{Otis I.B.}$

x	No.	$\bar{y}_p$	Y_p	$n_p (\bar{y}_p - Y_p)^2$
61-63	1	14.5000	22.7206	68
64-66	1	24.5000	28.9877	20
67-69	3	31.1667	35.2549	50
70-72	4	29.5000	41.5220	578
73-75	3	37.8333	47.7891	297
76-78	7	54.5000	54.0563	1
79-81	4	62.0000	60.3234	11
82-84	13	72.9615	66.5906	528
85-87	9	65.6111	72.8577	473
88-90	18	81.1667	79.1248	75
91-93	10	83.5000	85.3920	36
94-96	15	94.5000	91.6591	121
97-99	10	95.5000	97.9263	59
100-102	19	105.5526	104.1934	35
103-105	10	125.5000	110.4605	2262
106-108	12	106.1667	116.7277	1339
109-111	11	130.8636	122.9948	681
112-114	11	131.7727	129.2620	69
115-117	7	133.0714	135.5291	42
118-120	6	134.5000	141.7962	319
121-123	6	139.5000	148.0634	440
124-126	5	162.5000	154.3305	334
127-129	3	151.1667	160.5977	267
130-132	1	174.5000	166.8648	58
133-135	2	159.5000	173.1319	372
136-138	1	174.5000	179.3991	24
	192	—	—	8559

$$\bar{x} = 98.82813, \quad S(x - \bar{x})^2 = 45633, \quad \bar{y} = 99.65625, \quad S(y - \bar{y})^2 = 252795, \\ Sx(y - \bar{y}) = 95330, \quad S\{n_p(\bar{y}_p - \bar{y})^2\} = 207708, \quad b\{Sx(y - \bar{y})\} = 199149, \quad b = 2.089047.$$

Analysis of variance

Variation of Otis I.B.	Degrees of freedom	Sum of squares	Mean square	$\frac{1}{2} \log_e$
Linear regression	1	199149	—	—
Remainder between arrays	24	8559	356.6	2.9384
Within arrays	166	45087	271.6	2.8022
Total	191	252795	—	—

$$z = 0.1362$$

5% point approximately 0.229.

(3) *The distribution of Stanford-Binet Intelligence Quotients amongst Bath children*

It is now possible to estimate the mean and variance of Stanford-Binet I.Q.'s for the children composing the group.

The figures for the random group of 192 were

$$\text{Mean} = 98.708 \pm 1.121.$$

$$\text{Variance} = 241.433 \pm 24.706.$$

These children were, however, a random sample of 2553 children (the three oldest age groups) who were all tested on the Advanced Otis scale. The Otis I.B.'s for the 192 had a slightly lower mean and a greater variance than the 2553 I.B.'s from which they were drawn. Owing to the close association between the results of the two tests, it is possible to use this knowledge in order to secure improved estimates of the distribution of Stanford-Binet I.Q.'s. We are indebted to Prof. R. A. Fisher for his advice on suitable methods for making the adjustments.

Let $\bar{x}$ = mean Otis I.B. for the 192 children, $\bar{x}'$ = mean Otis I.B. for the 2553 children, $\bar{y}$ = mean Binet I.Q. for the 192 children, b = regression coefficient of Binet I.Q. on Otis I.B., and V = variance of the quantity following in brackets.

The adjusted mean I.Q. is simply the I.Q. at I.B. 100.081 (the mean for the 2553 children, as compared with 99.880 for the 192).

The sampling variance of the adjusted mean involves the addition of the usual term to the variance of the mean of I.Q.'s

$$(\bar{x}' - \bar{x})^2 V(b).$$

The value of $\bar{x}'$, however, is not known with accuracy, being an estimate based on 2553 results, so that an additional term must be added to the variance of the mean of I.Q.'s. This is:

$$b^2 V(\bar{x}').$$

The adjusted mean I.Q. thus becomes

$$98.785 \pm 1.152.$$

An adjusted value for the variance of I.Q.'s may be obtained by calculating

$$b^2 \sigma_x^2 + (\sigma_y^2 - b^2 \sigma_x^2).$$

The sampling variance of this estimate is

$$b^4 \frac{2\sigma_x^4}{2551} + 4b^2 \sigma_x^4 V(b) + \frac{2}{190} (\sigma_y^2 - b^2 \sigma_x^2)^2.$$

The adjusted variance thus becomes

$$231.065 \pm 15.388.$$

It will be noted that the mean I.Q. is close to 100, from which value it does not differ significantly. Similarly the variance is close to 225 (a standard deviation of 15), which appears to be accepted as the standard figure in much recent work (Thompson, 1932). The estimates for the standard deviation of Binet I.Q.'s of children included in the Scottish Survey (1933) were 17 for boys and 16 for girls.

(4) *Discussion and summary*

A random sample of 192 children was chosen by date of birth from the complete group and retested on the Stanford-Binet scale. It was composed of three subsamples that represented the range of chronological ages at Otis test. It was found that there was no

significant difference between the results for the subsamples, so it was possible to pool them and consider simply the entire group of 192. The regression coefficients were

$$\text{Binet I.Q. on Otis I.B.} = 0.3832 \pm 0.0145.$$

$$\text{Otis I.B. on Binet I.Q.} = 2.0546 \pm 0.0775.$$

Both regressions were linear. The correlation coefficient was 0.8873, the chance being 5 per cent that r was outside the range 0.8528 to 0.9141. The 1 per cent points were 0.8402 and 0.9211. There is thus a satisfactory and simple correspondence, particularly in view of the long interval between the two tests—about a year and a half in most cases.

The results given in the preceding portion of this paper permit some examination to be made of the norms we arrived at for the Advanced Otis scale (Table VIII in the first paper). The rate of increase of score with age was higher than that indicated by the norms in the Otis manual. A recent investigation by Gray & Moshinsky (1935), on 10,000 London school children, also gave a much lower value: a table of norms based on their results is given on p. 10 of the 1934 reprint of the *Otis Manual*. It should be noted that the effective size of the sample for the calculation of norms, only an incidental feature of their study, was far less than 10,000. Proportionately many times more secondary, private school children, etc., were included than were elementary school children. Thus, on weighting, there was a loss of some thousands of observations. Their norms were obviously based on a linear regression coefficient, as were ours for children up to the age of 12. The difference is considerable.

<i>Norms</i>		
	9 yrs. 6 mos.	12.0 yrs.
Gray & Moshinsky	61	99
Present study	52	98

While therefore the two sets of norms are very similar for older children, our figures are much lower for younger children. The preceding sections of the paper provide two lines of evidence that make it very unlikely that we over-estimated the regression.

In the first place the retest on the Stanford-Binet scale showed that an I.B. of 100 corresponded to an I.Q. of 101.1 in the youngest group, 98.5 in the middle group, and 96.6 in the oldest group. If, therefore, the Binet results were to be accepted as the criterion, our norms are actually too high for the younger children and too low for the older. It is quite possible, however, that this may be due, at least in part, to the age standardization of the Binet scale, rather than to that of the Otis scale. In any case, there is no evidence at all that our regression was too high.

In the second place, if our regression was accurate, the Otis retesting after a year showed an expected increase of 18 points, plus a practice effect of 9 points. This is a very high value for the practice effect after a year, so that again there is an indication that our regression may be somewhat too low.

In addition, there is no reason to suppose that our group of Bath children is in any way an unusual sample.

Gray & Moshinsky consider one criterion for assessing age correction, viz. the correlation between I.B. and age at test. This does provide a criterion for determining whether there is any evidence that norms obtained from one group of children are unsuitable for another group. Clearly, norms derived from a group cannot be tested on the same group, for the use of the regression in fixing the norms, and the use of the norms in determining I.B.'s, have already eliminated, as far as is possible, variability between arrays. Presumably therefore the correlations between age and I.B. given by Gray & Moshinsky in Table III in their paper, refer to I.B.'s determined on the American norms, and no independent evidence is given in their paper to support their own.

It is difficult to account for the difference between the results of the two investigations. One possibility may, perhaps, be very tentatively suggested. Our ascertainment was total, whereas that of Gray & Moshinsky was by schools. In the case of older children, these are to be found at secondary, central and senior schools for the most part; in fact, an efficient separation, in relation to mentality, has already been carried out. Also, senior schools are usually larger than junior schools, and the differences between them much smaller. Amongst children of junior school age, the difficulties of selection are far greater; the schools are smaller on the average and the differences between junior schools may be very great. The mean test results for different junior schools may differ by one, or even one and a half, times the standard deviation. It is clear, therefore, that it is a far harder task to secure a random sample of test results by schools, amongst children of junior school age and bias in selection will nearly always provide a sample that is superior to the general population.

There is one other factor of possibly considerable importance. A proportion of very clever children go on to secondary schools a year before the usual time. Clearly, if ascertainment is by schools, and weighting is proportional to the numbers at schools of the different types, this fraction will contribute disproportionately to the results for the younger children.

This hypothesis as to the difference is only put forward as a suggestion, and it may be that in the London survey, these sources of bias were in fact avoided. But whether or not a total ascertainment does in fact yield a better sample than does some other form of ascertainment, the results are undoubtedly much easier to interpret; a consideration that alone may to some extent justify the great amount of extra work involved.

We do not consider, therefore, that there is any evidence to show that our norms are seriously in error, while there is a good deal that suggests that they are reasonable values.

Utilizing the fact that the 192 children were a random sample of a large group of children, all tested on the Advanced Otis scale, the estimated mean Binet I.Q. was

$$98.785 \pm 1.152,$$

and the variance was

$$231.065 \pm 15.388.$$

This result is highly satisfactory. The mean I.Q. does not differ significantly from 100 and the variance is very close to the figure of 225 (standard deviation 15) that seems to be accepted as the standard figure in much recent work. There is no reason therefore to suppose that our group of Bath children differs in its mean level of I.Q., or in the variability of I.Q.'s, from what appear to be reasonable estimates for the general population in this or other countries.

NOTES ON THE USE OF A SHORTENED BINET SCALE

In the preceding section the results refer to the full Binet scale as there defined. More than one method of shortening the scale has been employed, and as very many individual tests have to be carried out during our investigations, it was thought desirable to make an estimate of the error introduced by the use of one such shortened method.

In the shortened scale that we are using, if a child passes all but one of a complete year of tests it is credited with passes in all tests below that year. If it fails all but one of a complete year of tests, it is assumed to have failed all tests above that year. In no case would less than 3 years of tests be given. Also, if a child were to pass all but one in 1 year and fail all but one in the next year above, the scale would in that case be continued both upwards and downwards for at least 1 year.

In order to determine the effect of shortening the scale in this way, it was possible to analyse the detailed results for the 192 random children tested on the full scale (less two for whom detailed forms were not available) as described in the preceding section. In each case a new result was worked out on the basis of the shortened scale, when that was applicable. These children were nearly all 11-14 years old at Binet test and it was desirable to extend the analysis by including younger children. At the time that the calculations were made, 122 full Binet results were available for children ranging in age from 4 to 10; these were brothers and sisters of children in the main group, selected for test for a variety of reasons. As regards the distribution of I.Q.'s, the younger group proved to be almost identical with the random group.

	Mean	Variance
Random group	98.836	238.772
Younger group	99.557	238.761

As the analysis proceeded, it became evident that in every other respect, as far as we could discover, there was no appreciable difference between the two groups as regards the relation of the shortened to the full scale. Not only, therefore, are the conclusions about the use of the shortened scale valid for a very wide age range, but the more adequate number of 312 results, apparently forming a random sample, could be subjected to the necessary examination.

In its simplest form the analysis falls into two parts. First, what proportion of children

pass all but one of a complete year of tests, and of these how many will fail in one, two or more tests if the scale is extended downwards? Similarly, in the second place, how many children fail all but one of a year of tests, and of these how many will pass one, two or more additional tests if the scale is extended upwards?

In both cases the results did not differ appreciably for the random and younger groups, so they are taken together. Taking first the extension of the scale downwards, the results are shown in Table XVIII.

Table XVIII

	No.	Percentage
Cases in which downward extension did not apply	181	58.0
Cases in which downward extension did apply	131	42.0
	<u>312</u>	

Additional tests failed on downward extension.

$$m = 0.4046$$

No. of tests failed	Observed frequency	Expected frequency	χ^2
0	88	87.45	0.003
1	34	35.38	0.054
2	8	7.16	0.070
3	1	0.97	
4	0	0.10	
5	0	0.01	
	131	—	0.127

$n = 1$ and P lies between 0.70 and 0.80.

The distribution of additional failures amongst children to whom the shortening was applicable reproduces a Poisson series very accurately.

$$e^{-m} \left(1, m, \frac{m^2}{2!} \dots \frac{m^x}{x!} \right)$$

where m is the mean frequency of occurrence. The fitting is shown in the Table and the discrepancies are not significant.

The results of the upward extension of the scale are shown in Table XIX. It will be noted that the distribution is not a regular Poisson series in this case. This is due to a few children who were able to pass higher tests of a practical type, particularly those involving the repetition of numbers, up to a far higher level than the rest of their performance. For example, it is possible for a child to pass such tests up to the Superior Adult level, although it does not pass any other test above the 12-year level. But for this peculiarity of the Binet scale, which affects only a small proportion of children, it is not unlikely that a Poisson series would be applicable in the upward direction also.

Owing, however, to the complexities of the manner in which I.Q.'s are derived, there is no way of using this information directly, as long as it is I.Q.'s that are being considered. First of all, a single test has a value of 2 months' mental age up to year 10, of 3 months' at year 12, of 4 months' at year 14, of 5 months' at Average Adult level, and of 6 months'

at Superior Adult level. Except, therefore, in the case of young children, a discrepancy in the upward direction will have more effect upon the I.Q. than one in the downward. Further, as the divisor in determining I.Q. is the chronological age, the effect of a discrepancy will fall with age, except that this is balanced to some extent in the upward direction, because young children show discrepancies at the levels at which the value of a single test is small.

Table XIX

	No.	Percentage
Cases in which upward extension did not apply	190	60.9
Cases in which upward extension did apply	122	39.1
	312	

Additional tests passed on upward extension.

$$m = 0.3443$$

No. of tests passed	Observed frequency	Expected frequency	χ^2
0	93	86.46	0.495
1	21	29.77	2.584
2	3	5.12	0.871
3	5	0.59	
4	0	0.05	
	122	—	3.950

$n = 1$ and P lies between 0.05 and 0.02.

As the preceding sections of this paper deal with Binet results entirely in the form of I.Q.'s, it is perhaps best to examine the effect of using the shortened scale upon the I.Q., however inadequately this may represent the true underlying relationship shown in the preceding paragraphs.

The distribution of discrepancies in I.Q. cannot be expected to show a normal, or indeed any other regular arrangement; nevertheless it will be seen that for purposes such as ours, it can be shown that the error introduced by the use of the shortened scale is so small that it can be ignored.

The distribution of differences in I.Q. is shown in Table XX. It will of course be realized that in a proportion of cases the scale will be shortened in both directions, and so differences will entirely or partially cancel each other. Again the random and younger groups did not differ appreciably. The mean deviation did not differ significantly from zero, so that there is no evidence that plus and minus differences do not cancel each other out on the average, if groups are being considered. The large differences in one direction only are of course due to the high value in mental age of tests at the higher levels, when there is no corresponding difference in the other direction in those particular cases.

The association between I.Q.'s derived from the full and shortened scales is shown in Table XXI. Again the difference between random and younger groups was very small. For example, if the difference in correlation coefficients is examined, it will be seen that it is not significant.

Table XX. *Difference between I.Q.'s on full and shortened scale*

+ indicates that full I.Q. is greater than shortened I.Q.
 - indicates that it is smaller.

Difference	Random group	Younger group	Total
+ 11	1	0	1
+ 10	1	0	1
+ 8	1	1	2
+ 7	0	1	1
+ 6	0	2	2
+ 5	1	3	4
+ 4	3	2	5
+ 3	7	1	8
+ 2	4	1	5
+ 1	0	0	0
0	148	96	244
- 1	10	1	11
- 2	11	9	20
- 3	2	4	6
- 4	1	1	2
	190	122	312

Table XXI. *Association between I.Q.'s on full and shortened scales*

x = full I.Q.; y = shortened I.Q.

Random group

$n = 190$, $\bar{x} = 98.83684$, $S(x - \bar{x})^2 = 45127.9$,
 $\bar{y} = 98.66316$, $S(y - \bar{y})^2 = 44054.4$, $Sx(y - \bar{y}) = 44331.6$.

Regression of shortened I.Q. on full I.Q. = 0.982353.

Regression of full I.Q. on shortened I.Q. = 1.006290.

Correlation coefficient = 0.994250.

Younger group

$n = 122$, $\bar{x} = 99.55738$, $S(x - \bar{x})^2 = 28890.1$,
 $\bar{y} = 99.40984$, $S(y - \bar{y})^2 = 28203.5$, $Sx(y - \bar{y}) = 28343.1$.

Regression of shortened I.Q. on full I.Q. = 0.981067.

Regression of full I.Q. on shortened I.Q. = 1.004951.

Correlation coefficient = 0.992937.

Total

$n = 312$, $\bar{x} = 99.11859$, $S(x - \bar{x})^2 = 74056.6$,
 $\bar{y} = 98.95513$, $S(y - \bar{y})^2 = 72299.4$, $Sx(y - \bar{y}) = 72714.7$.

Regression of shortened I.Q. on full I.Q. = 0.981880.

Regression of full I.Q. on shortened I.Q. = 1.005744.

Correlation coefficient = 0.993740.

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	r	z	$n'-3$	$\frac{1}{n'-3}$
Random	0.99425	2.9245	187	0.005348
Younger	0.99294	2.8214	119	0.008403
Difference	—	0.1031 ±	0.1173	0.013751

One more comparison may be given to illustrate the small effect of age differences on the results. Dividing the younger group into two, the mean square difference between the pairs of I.Q.'s derived by the two methods was 3.51 for 50 children ranging in age from 4 to 7 inclusive, and 3.13 for 72 children aged 8 and 9 years.

Taking the figures in Table XXI at their face value, the error variance due to the use of the shortened scale is so small that it may be ignored as being negligible in comparison with other errors inherent in studies such as this. Actually, however, the error variance has been increased by the high outlying plus values, and also big discrepancies are more numerous than they should be, if the distribution of differences were normal. It is, however, very arguable that, in these cases of big discrepancies due to an ability to succeed at such tasks as repeating numbers, the shortened scale actually gives the better estimate. Taking all these considerations together, it seems clear that the use of the shortened scale is fully justified when used in ordinary statistical studies, and that any adjustment that might be made in consequence is too trivial to be worth while using.

The extra years of tests necessary for the 312 children, if the full Binet scale were to be used, as compared with the shortened scale, was 329. On the average, therefore, 1.05 years of tests per child would be saved by using the shortened scale. A moderate estimate for the time saved per child would be 8 min., a very considerable economy.

GENERAL SUMMARY

1. In the previous paper of this series a group of 3400 school children was defined as all children whose homes were within the boundaries of the City of Bath on 27 July 1934, and whose dates of birth fell between 1 September 1921 and 31 August 1925 inclusive. These children have been tested on the Advanced Otis scale.

2. In the present paper is described a retest of 532 children on the same scale after the lapse of 1 year. The subsamples were composed of children drawn from two age groups that represented the extreme difference in age at original test. Both were representative as regards score at first test.

3. The correlation coefficient of original and second scores was +0.9473 (for both groups together). Both regressions were linear.

4. The method of equalization for age differences, elaborated in the first paper, appeared to work reasonably well. Some indirect evidence is given that supports the values in our table of norms.

5. The increment during 1 year was independent of first score, as was also the square of the increment. Within each age group the increment was also independent of age at first test, but a comparison of the two subsamples, widely separated in age, showed that the older children added fewer marks during a year. There was a definite practice effect.

6. A random sample of 192 children was retested on the Stanford-Binet scale, in most cases after an interval of more than a year. The correlation coefficient of Otis I.B.'s and Binet I.Q.'s was 0.8873. Both regressions were linear. The range of ages at test within the group had no appreciable effect upon the relationship.

7. The distribution of Binet I.Q.'s for the Bath children was very close to the generally accepted standards for an unselected population, i.e. mean 100 and variance 225 (standard deviation 15).

8. The use of a shortened Binet scale has been investigated for 312 children, who apparently formed a good random sample. Subject to certain limitations, if a child passes all but one of a year of tests, it is credited with passes in all years below. The scale is shortened similarly in the upward direction. The distribution of additional failures in lower years, amongst children who pass all but one of a year of tests, conforms to a Poisson series. In the upward direction there is a divergence from a Poisson series, due to disproportionate ability to succeed at mechanical tasks. The magnitude of the effect upon the I.Q. of using the shortened scale is demonstrated and is shown to be very small. The relationship between the full and shortened scales does not vary appreciably over a wide range of age. The time saved is about 8 min. per child.

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APPENDIX TABLE III
Retest on Stanford-Binet scale

Binet I.Q.	Otis I.B.		
	First group	Second group	Third group
61	—	12	—
65	—	21	—
67	36	—	—
68	—	28	30
70	28	—	29
71	—	—	37
72	34	—	—
74	28	48	—
75	48	—	—
76	50	—	—
77	62	60	46, 53
78	22	74	—
81	82	56	58, 58
82	84	56, 62, 79, 83	79
83	64, 77, 82	45, 86	—
84	92	67	—
85	91	—	58, 70
86	65, 75, 76	33	54
87	68	—	—
88	—	61, 67, 75, 79, 88, 97	68
89	80, 110	—	57, 65, 83, 91, 106
90	—	96	55, 96, 96
91	72, 91, 129	74, 77, 84	—
92	92, 98	—	59, 66
94	115	75, 90	79, 91, 95, 96
95	80	98, 101	—
96	96, 108, 146	—	45, 108
97	118, 121	102	56, 89, 105
99	106	94, 95	76
100	79	90, 99, 118	84, 99, 104, 105
101	103, 113, 118, 119, 141	97, 130	113
102	—	110	89, 102
103	115, 131, 139	110, 126	137
104	—	119	129, 131
105	—	129	—
106	—	110	89, 119
107	102	135	96
108	91, 111, 115	—	82, 98, 126
109	134	—	165
110	113, 124, 129, 140	150	106, 119, 127, 139
112	145	161	—
113	147	129, 134, 145	115
114	126, 132	—	102, 114
115	142, 157	—	112, 121
116	126, 138	132	—
118	—	137	125, 138
119	—	133, 144	—
120	—	130	—
121	—	—	134
122	—	—	133
123	149	132, 136	151
125	—	150	—
126	142, 164	162	181
127	172	126	—
128	—	—	150
131	—	170	—
133	—	—	166
135	156	—	—
138	—	—	176

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